30/10/25

6. Retrival Augmented Generation

1. Deploying and Maintaing RAG system
1. Building and Deploying RAG in Production.

① Understanding RAG system →

→ LLM are revolutionizing industry and is helping in creating vast opportunities.

→ One of the critical usage of LLM is RAG

→ Prompt Engineering is not sufficient.

Eg

Input
Prompt → LLM → Response

→ But this usage pattern is insufficient for many cases in Enterprises.

→ One approach to solve the problem is to stuff Everything in prompt →

→ Input
prompt → LLM → Response
Context

stuff all the information here.

→ Limitation of using context window →

(i) Limited context window

(ii) Quality degradation → (as stuffing too much may increase noise that leads to degradation)

→ common Approach to solve the problem is RAG.

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→ RAG →

→ RAG consists of two pipelines → ingestion pipeline and RAG pipeline

→ Ingestion pipelines → run offline

Data preparation → Enterprise data → chunks

Indexing → Embedding model → (vectordb) (that stores numerical Embedding of chunks)

chunks → chunks to be stored in vector db as embeddings

→ After calculating all the Embedding of each chunks they are stored in vectordb.

→ Rag Pipelines → when we input a query that query is ~~converted~~ converted into Embeddings ~~by~~ using the same embedding models ~~to get~~ and stored in the vectordb as query embeddings

→ Then query embeddings are compared with the chunks embeddings to get the relevant similarities

(1) Reterival phase → to get the top relevant chunk or embedding comes under ~~Retrev~~ Retrieval phase of Rag pipeline.

(2) Augmented phase → with the retreived chunks and prompt template the data is sent to the LLM model to ~~get the~~ generate the response

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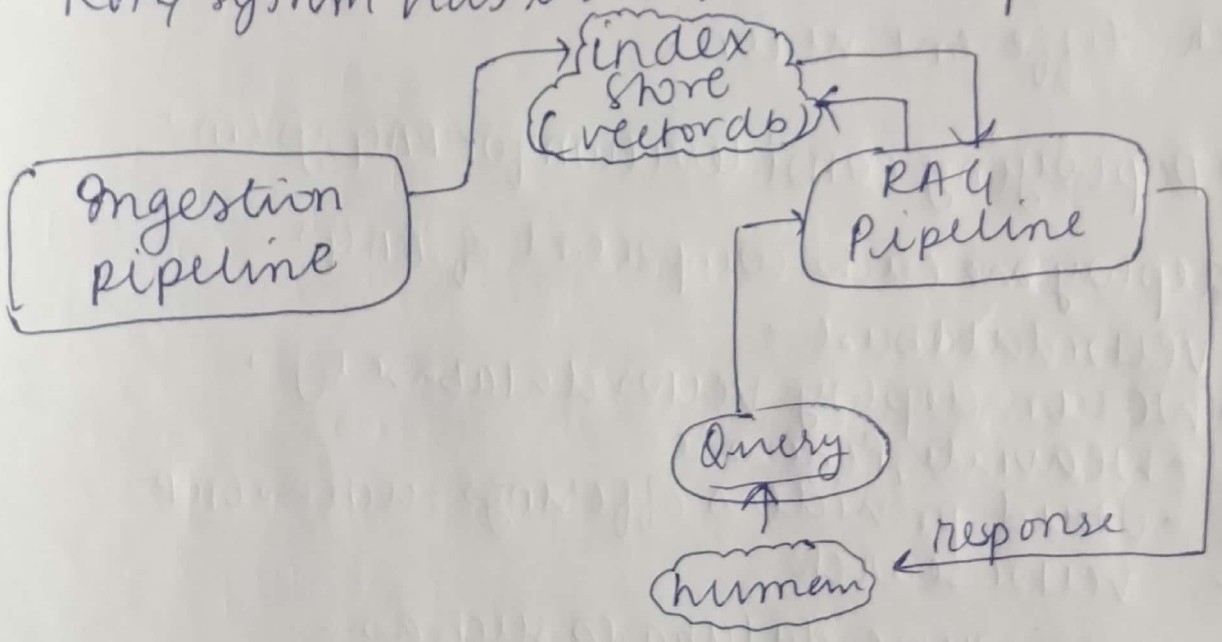
→ Benefits of RAG systems →

1. Include your private and recent data
2. Low cost based on usage as it is used only for one time.
3. Grounding and reducing hallucination as the relevant chunks or embedding are retrieved that reduces hallucination.
4. source citation and attribution

→ RAG system has three main components →



→ choices to be made while building RAG system.

1. Parsing and chunking strategy
2. Embedding mode and configuration
3. Retrieval techniques
4. Top N chunks
5. LLM and associated parameters
6. Prompt template

→ To make the construction efficient we can use frameworks (RAG frameworks), like Langchain, Llamaindex

V2 → Demo Building simple RAG system →

→ Creating a RAG system by using llamaIndex
→ Usecase.
→ Online Retailer company → Foo (adventure gear)
→ Founded by → Bar (person name) (CEO)
→ Quax (co-founder with Bar)

with different vision -

V3 → Deploying consideration for RAG system →

To deploy each component of RAG →

① vector database →
→ we can deploy vector database by →
Managed db { - Weaviate, Pinecone
- cloud providers offerings such as GCP vector search

self hosted and managed db { - Open-source databases such as -
(a) chroma,
(b) milvus
(c) elasticsearch

→ and these are deployed as containers with persistent volumes
→ and that can be further orchestrated by using popular frameworks like Kubernetes

② Ingestion Pipeline →
→ For Ingestion Pipeline, we can take ingestion pipeline as code base and can run as job
→ job → either one time or scheduled at regular intervals.

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→ Or we can deploy the Ingestion pipeline as API

→ we can also add advanced features like (3) triggering the Ingestion Pipeline if a new doc is created or when existing document get updated, ~~and~~

→ Embedding model used in the ingestion process ~~pro~~ can be either commercial APIs such as OpenAI or Bedrock or can hosted and managed on infrastructure.

(3) RAY Pipeline →

→ for RAY Pipeline as well we can wrap it as a ~~pip~~ API and deployed as containers and Orchestrated by kubernetes.

→ for Embedding & llm model → either chose an commercial API or self-hosted model.

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1.2 Managing RAG

(2) (V1) → managing functional Requirements Metrics

→ Managing RAG Ingestion Pipeline
→ Need to manage →
(1) Document lifecycle → like addition, updation or removal of document.
[→ Ingestion pipeline must make sure that vector database contains the most upto-date information.]
(2) Parsing and chunking strategy → need to adjust the document parsing and chunking based on the type of document.
For ex → Text documents requires diff. parsing and chunking strategy as compared to images doc.
(3) embedding models → need to try diff. embedding models., including industry tailored models to capture the semantic meaning.

→ In RAG pipeline, we need mechanisms to measure and monitor several key functional metrics to ensure the efficiency of RAG pipeline created.

→ RAG Pipeline Evaluation Metrics →
(1) Completeness →
→ It refers to check that the response generated by the rag pipeline addresses all the part of the query. And our goal is to increase completion score.

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② Faithfulness →

→ This metric measures the factual consistency of the generated response with the retrieved context.

③ Toxicity (safety metric) →

→ It is a safety metric that evaluates RAG pipeline against various toxicity definitions such as personal attacks, hate or dismissive statements

④ Bias →

→ Bias measures the response on the basis of Biases such as → gender, political racial or geographical bias in the generated response.

• To measure these functional metric we can use RAG Evaluation frameworks and tools →

① Deepeval
② Ragas
③ Truleus
④ Galileo

→ for Enterprise grade rag system apart from functional metrics we have to consider other things as well such as →

① Access Control →

→ since in Rag system we will be using multiple documents and have different knowledges bases, it is important to ensure that the users are authorised. and all the authorised people can access the information only on the basis of role and profile.

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② Security →

→ We need to implement appropriate security measures for the data ingestion ~~path~~ rag pipelines and API security to prevent →

① Unauthorized access

→ and ensures →

① Encrypted communication such as https.

→ and establishs →

① Guardrails.

Attacks →

- DDOS
- Prompt Injection

③ Latency →

→ Latency means the difference b/w time of query and the final response

→ Minimising Latency is essential for better user experience.

→ Monitoring and Reporting →

→ Different metrics

- Functional KPIs
- Non-functional KPIs

→ Compliance →

- Legal and regulatory compliance
- Audit reports
- Risk mitigation (governance mechanism helps in mitigating risks)

KPIs → Key Performance Indicator

KPIs → measurable value that demonstrates how effectively a company, department or individual is achieving key business objectives

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V3 → Going Beyond simple RAG system:

→ Simple RAG has limitations

→ For complex RAG system:

→ We can ~~ing~~ improve ingestion Pipeline by:

1. utilising various data parsing and chunking strategy
2. Or by using different embedding model.

→ Limitations of Naive RAG pipelines:

1. Naive RAG pipelines lacks memory
2. Naive RAG lacks summarization capacity (as it only enables the semantic search based on the ~~re~~ retrieved top chunks)

→ For complex RAG system we need:

1. memory for conversation
2. summarisation → we need all the chunked to be processed
3. Adding Routing capability → is used to utilise the rag system for various tasks such as semantic search or summarization based on the input request.

→ We can improve the complex RAG to the next level by creating Agentic RAG :-

→ That helps in →

1. Multiples tools and integrations
2. enabling multi-turn question
3. complex reasoning.
4. Performing self-reflections and correction.

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